A-30

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Roll No. .....

D028811(028)

B. Tech. (Eighth Semester) Examination, Apr.-May 2023

(Electronics and Telecommunication Engg. Branch)

(AICTE Scheme)

ADVANCED COMMUNICATION

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note *: All questions are compulsory. Part (a) of each unit is compulsory and carries 4 marks. Attempt any two parts from (b), (c) and (d) and carries 8 marks each.*

Unit-I

1. (a) Explain Launching process of Communication Satellite. 4

(b) Describe about the GPS, VSAT and DTH of satellite communication system. 8

(c) Write the name of different orbital parameter and define them at least four. 8

(d) State the Kepler's Laws and find out the height of satelite in case of geosynchronous satellite system. 8

Unit-II

2. (a) Explain why it is necessary to establish the satellite link. 4

(b) A 36 MHz BW transponder has been rented by user. The user's earth station achieves overal CNR of 15 dB. Determine the maximum numbers of channel that user can operate in FDM/FM/FDMA system to achieve worst channel weighted SNR of 51 dB. 8

(c) Derive the expression for received power for satellite communication. 8

(d) Derive the expression for SNR of FM Detection. 8

Unit-III

3. (a) What is the difference between FDM and OFDM techniques. 4

(b) Give the basic details of Intelsat TDMA system. 8

(c) Write the principle of OFDM and implementation of OFDM transceivers. 8

(d) The five Earth station all operate in TDMA mode. Transponder BW is 36 MHz. The original speech signals are sampled at 8 kHz rate 8 bits per sample. System uses QPSK. Find the bit rate for each PCM signal? What is the frame time for TDMA without storage and what is the maximum rate of transmission that can be used and how many telphonic channels can be transmitted? 8

Unit-IV

4. (a) Define and briefly explain TIR and NA. 4

(b) Define Attenuation and explain the mechnaism of attenuation. 8

(c) Explain the working of SMSIF, MMSIF, SMGIF and MMGIF with necessary diagrams. 8

(d) For a silica fiber refractive indices are 1.48 and 1.46 for core and cladding respectively and for another glass fiber refractive indices are 1.495 and 1.402 for core and cladding respectively now find out acceptance angles and numerical aperture for these fiber cables. 8

Unit-V

5. (a) What is the difference in LED and LASER.

(b) Draw the block diagrams for SLED and ELED used for optical communication and explain their working.

(c) Explain the working of photo diode. Discuss the application of it.

(d) Explain Population Inversion and mechanism of photon emission in Laser Diodes.